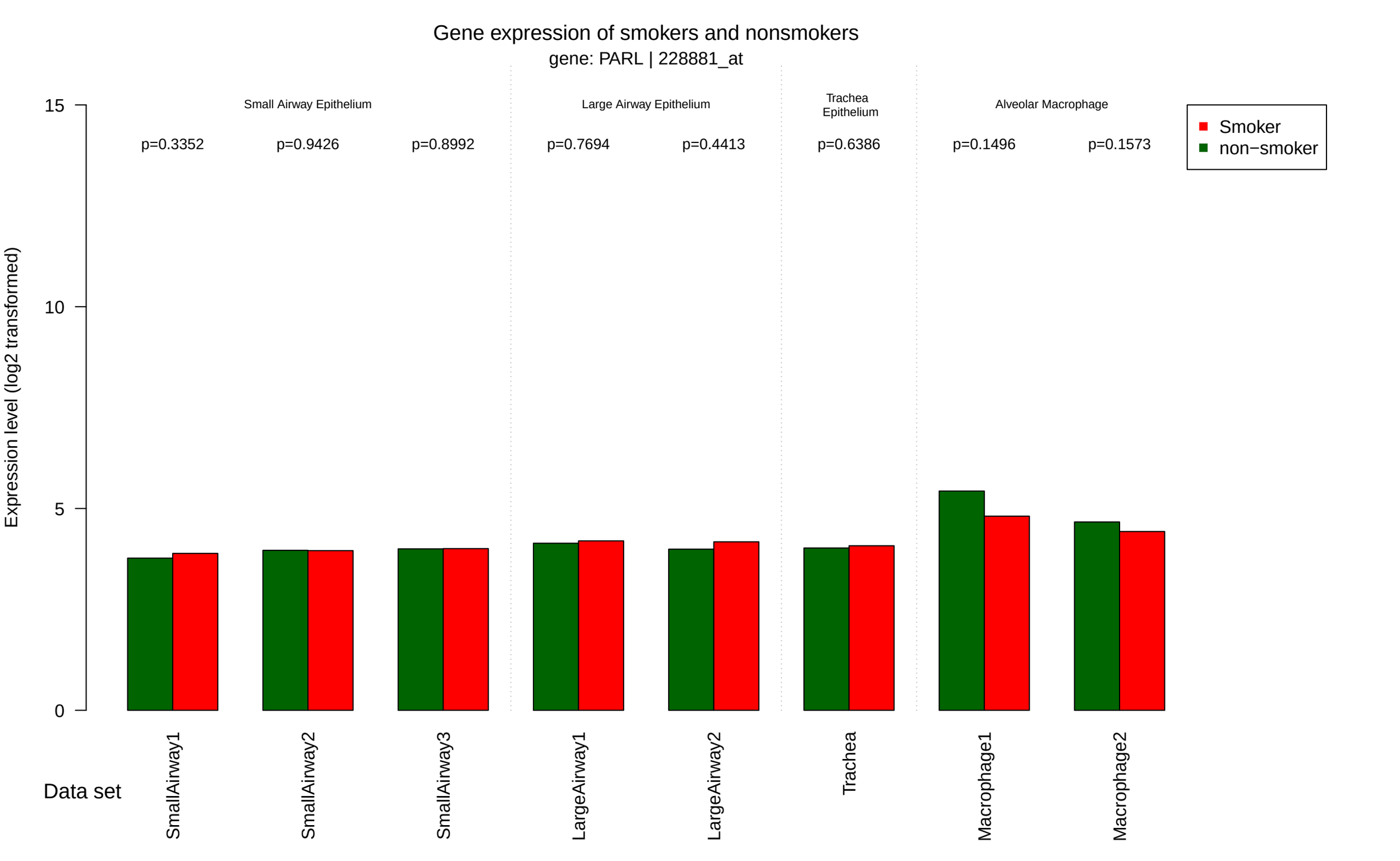
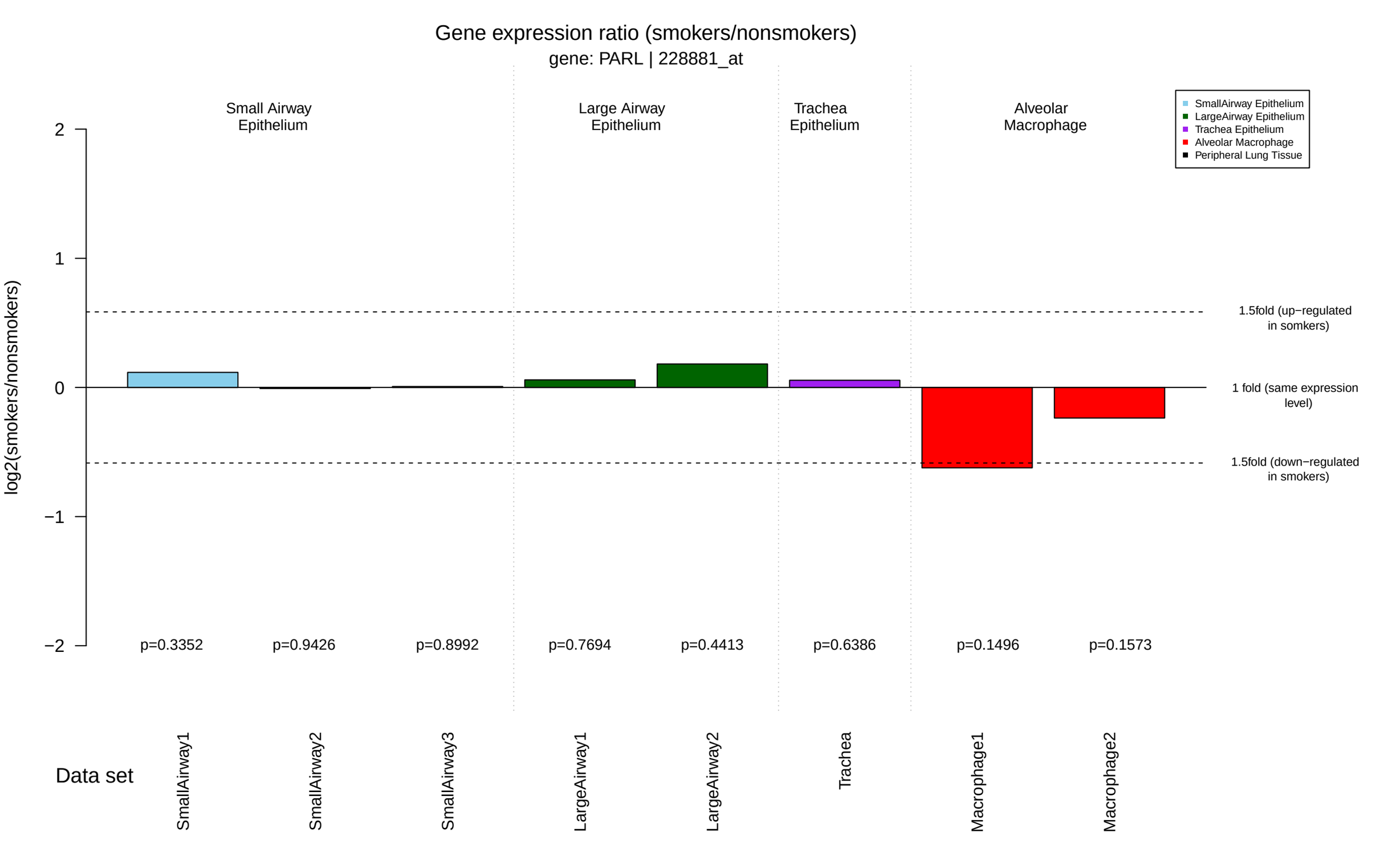
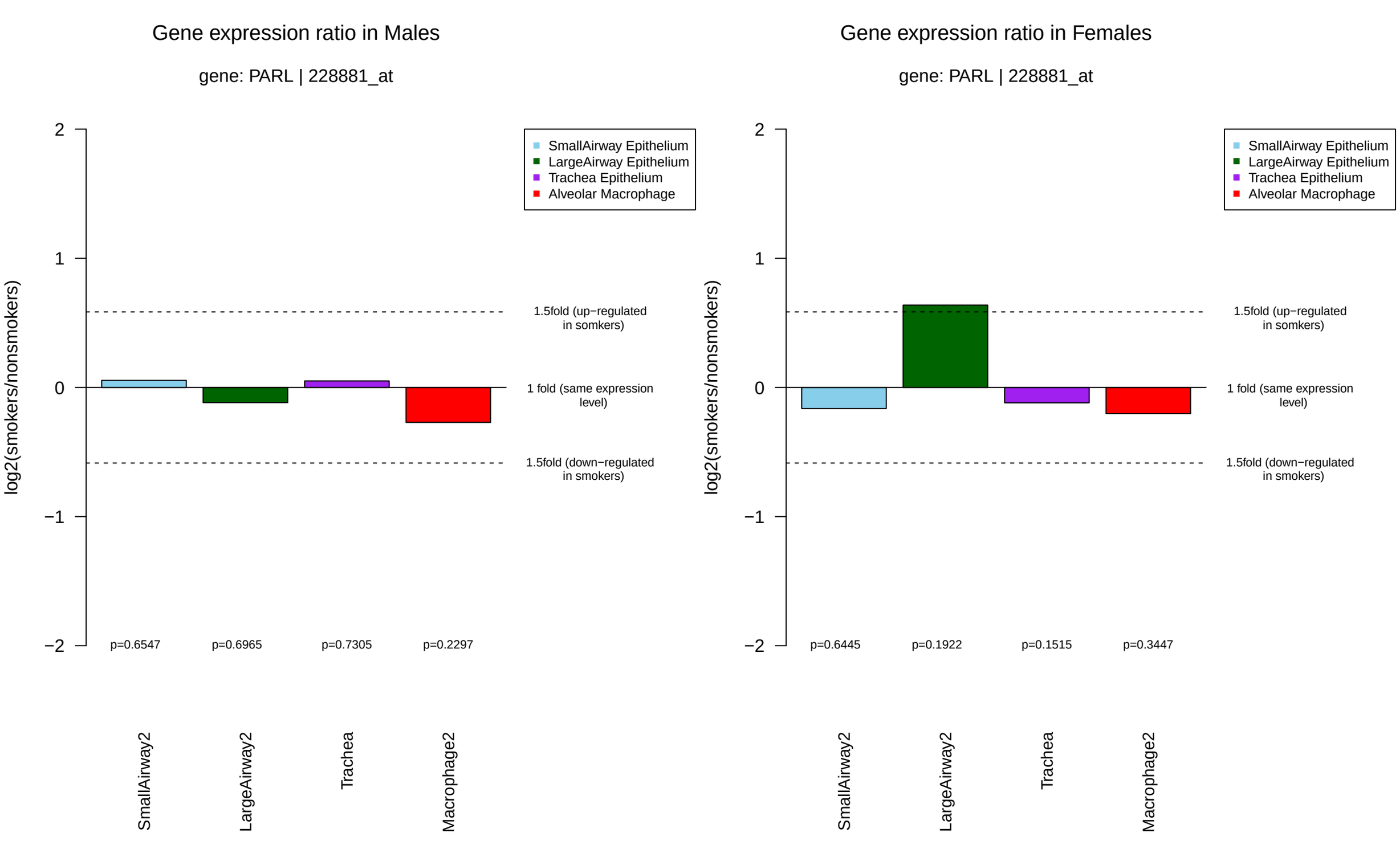
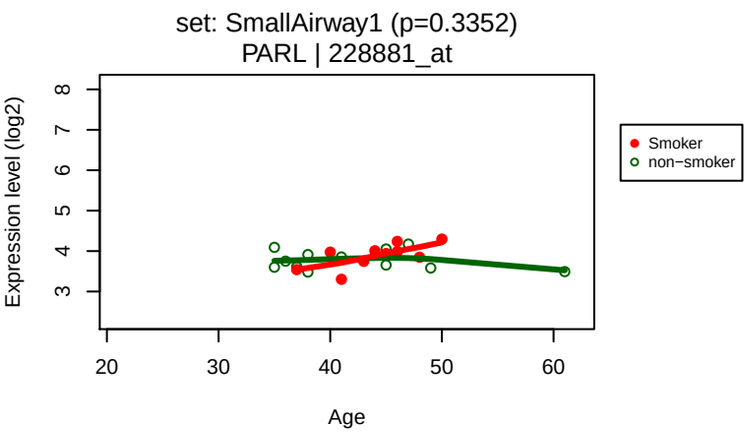
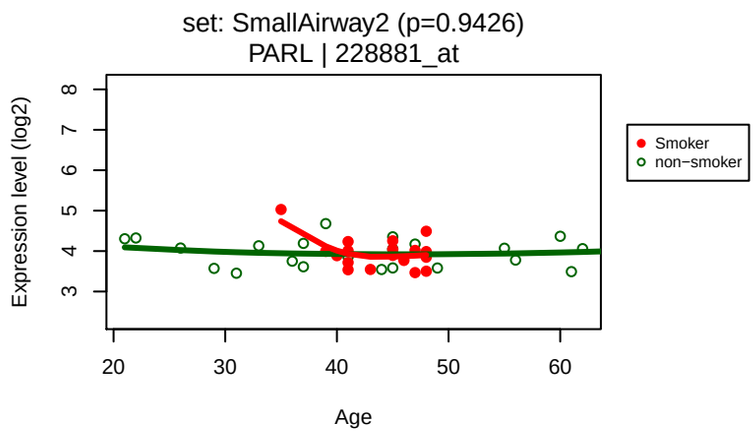
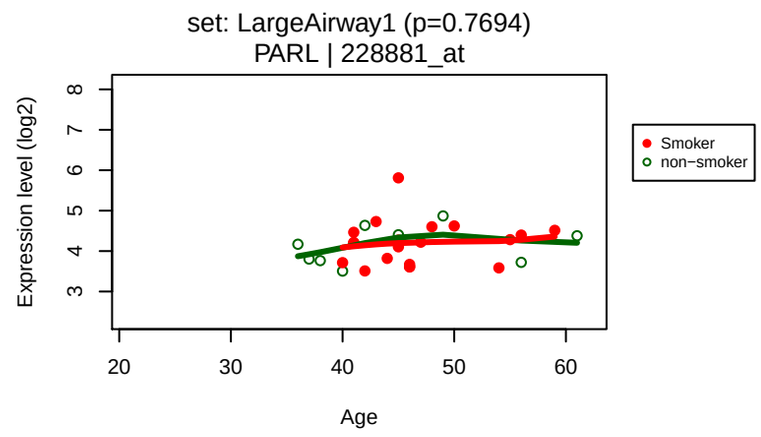

Gene expression of smokers and nonsmokers
gene: PARL | 228881_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3352
p=0.9426
p=0.8992
p=0.7694
p=0.4413
p=0.6386
p=0.1496
p=0.1573
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

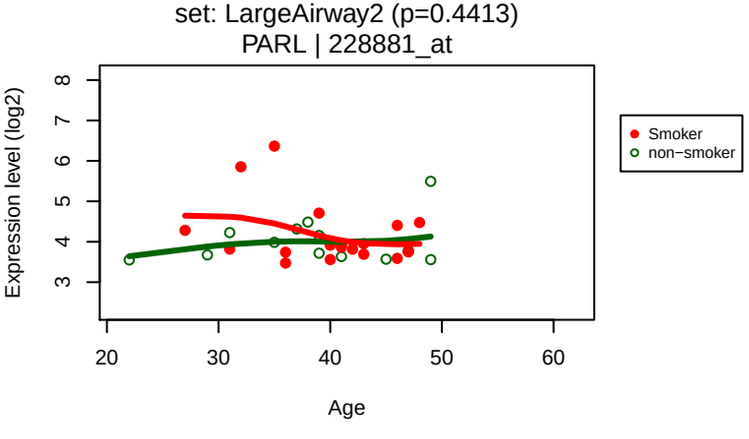

Gene expression ratio (smokers/nonsmokers)
gene: PARL | 228881_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3352
p=0.9426
p=0.8992
p=0.7694
p=0.4413
p=0.6386
p=0.1496
p=0.1573
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

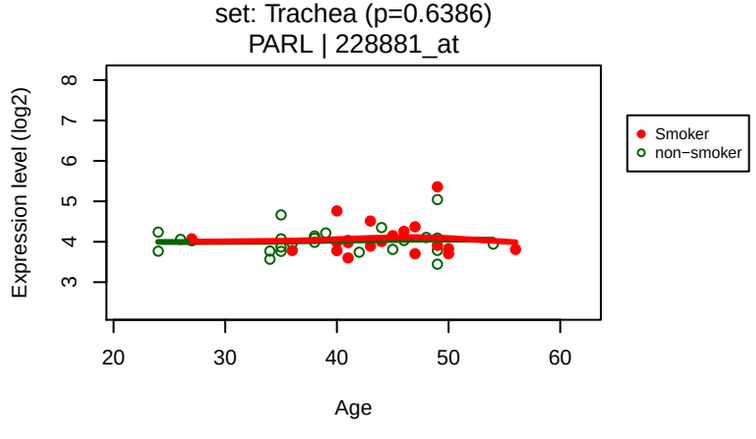

Gene expression ratio in Males
gene: PARL | 228881_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6547
p=0.6965
p=0.7305
p=0.2297
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PARL | 228881_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6445
p=0.1922
p=0.1515
p=0.3447
SmallAirway2
LargeAirway2
Trachea
Macrophage2

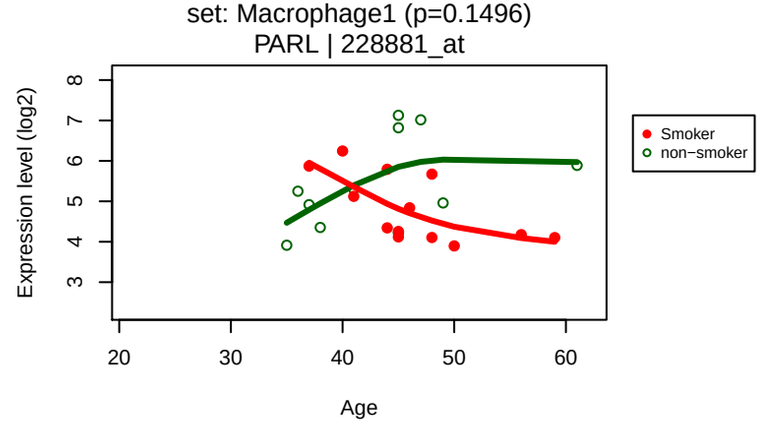

set: SmallAirway1 (p=0.3352)
PARL | 228881_at
Expression level (log2)
Age
Smoker
non−smoker

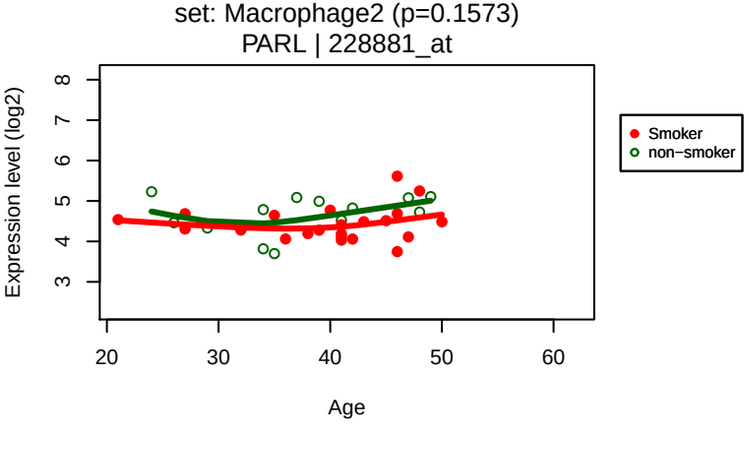

set: SmallAirway2 (p=0.9426)
PARL | 228881_at
Expression level (log2)
Age
Smoker
non−smoker

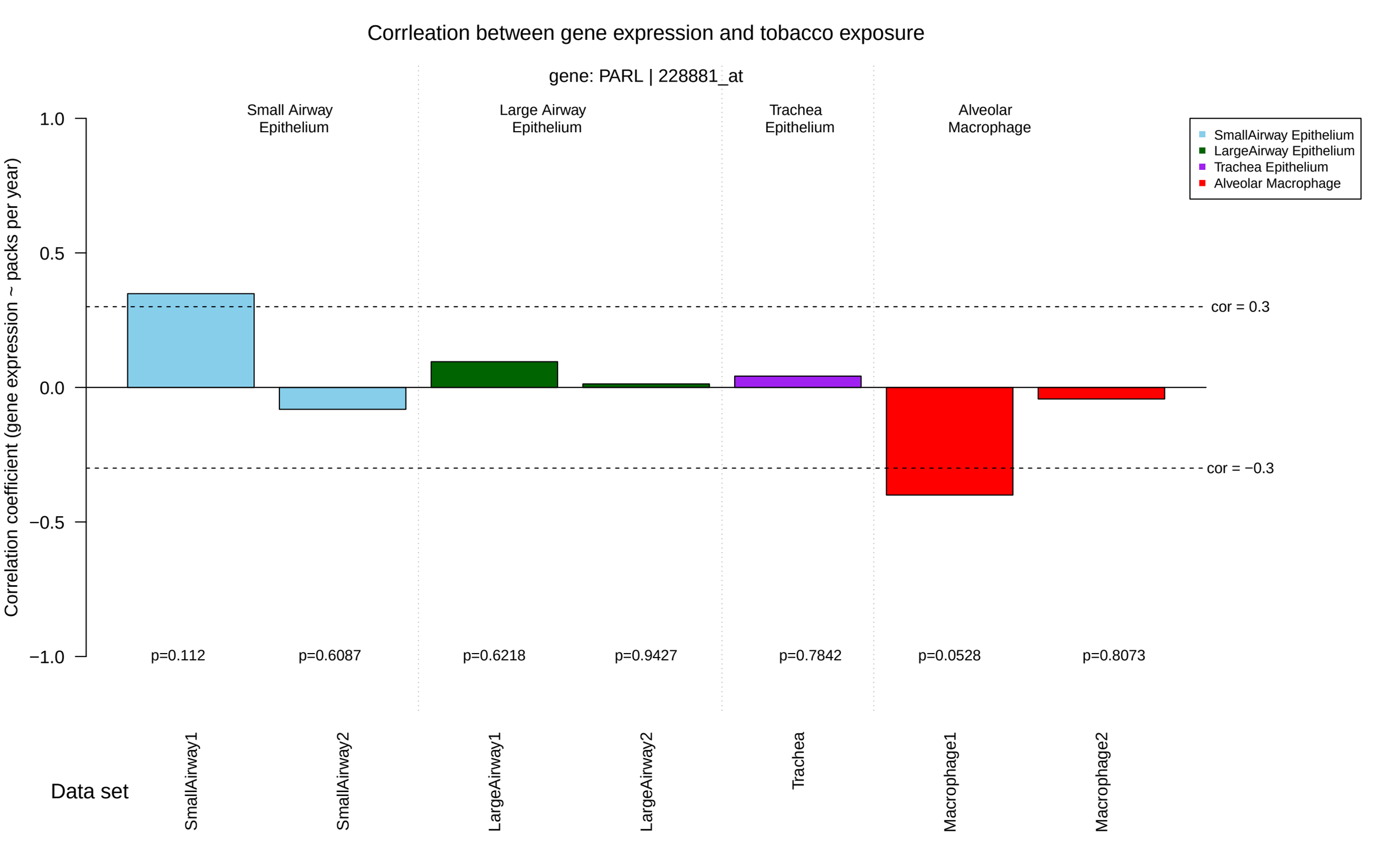

set: LargeAirway1 (p=0.7694)
PARL | 228881_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4413)
PARL | 228881_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6386)
PARL | 228881_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1496)
PARL | 228881_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1573)
PARL | 228881_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PARL | 228881_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.112
p=0.6087
p=0.6218
p=0.9427
p=0.7842
p=0.0528
p=0.8073
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2